

Bench Heat and Wetness in Central Park, June to November

Assignment 3 · Ishika Agrawal

1. What I'm attempting

I want to estimate how many hours each of Central Park's 477 mapped benches is comfortable to sit on between June and November. In summer, benches in direct sun get hot. In fall, rain and dew leave benches wet, and shaded ones take longer to dry. I'll run the benches hour by hour through one season, then repeat it with the warmer summers expected in the 2050s. This lets me compare open and shaded benches across the season and see how warming changes the number of usable hours.

2. The dataset

OpenStreetMap (OSM), Central Park benches, downloaded in September 2026. OSM is a free map built by volunteers since 2004 from GPS traces, aerial photos and local knowledge, mainly for navigation. It has 477 rows, one per bench. The columns I would use are the bench ID, its location (in feet, on the New York map grid), and its material (mostly left blank). I would add three other sources:

- **Tree cover** over each bench (yes/no), from NYC's 2017 land cover map.
- **Weather** from NOAA's Central Park station: hourly temperature (°F), dew point (°F), rainfall (inches) and wind (mph).
- **Sunlight** for each hour, from the National Renewable Energy Laboratory's solar database.

3. Research

- **Heat.** Vanos et al. (2015, Arizona; 2022, *Building and Environment*) measured how hot playground surfaces get in sun and shade, across materials, and when they become hot enough to burn skin. They found shade and material matter more than air temperature. This is the basis for my heat rule.
- **Wetness.** Plant scientists measure how long leaves stay wet after rain or dew, because it predicts crop disease. Sentelhas et al. estimate this from ordinary weather station data. I apply the same idea to bench slats.
- **Future climate.** The New York City Panel on Climate Change publishes expected temperatures for the 2050s.

4. My record

477 benches, each with a location, tree cover and material (I assume wood where it's blank), plus the hourly weather in Central Park from June 1 to November 30, 2025.

5. My rule

Every hour, each bench is checked for three things:

- **Heat:** the bench warms up with the sun hitting it. Shaded benches stay close to the air temperature; benches in full sun get much hotter. Above a set temperature, it's too hot to sit on.
- **Getting wet:** the bench gets wet if it rains that hour, or if dew forms overnight.
- **Drying:** a wet bench dries faster in sun and wind, and slower in shade.

A bench counts as **sittable** when it's neither too hot nor wet.

Number the rule needs	Where it comes from
Hourly temperature, dew point, rain, wind	NOAA Central Park weather station
Hourly sunlight	National Renewable Energy Laboratory
How much shade a bench gets	Me, from the tree cover map
How fast a bench heats up in sun	Me, based on Vanos's measurements (least sure)
How hot is too hot to sit (about 120°F)	Me, set below Vanos's burn levels (least sure)
How fast a bench dries	Me, adapted from the leaf wetness studies (least sure)
When dew forms	Leaf wetness studies

6. My run

- **Step:** one hour.
- **Span:** June 1 to November 30, about 4,400 hours, run twice.
- **2025:** the actual weather from that year. Nothing is left to chance.
- **2050s:** the same weather made warmer by the expected amount. Since that amount is uncertain, I run it 100 times, each with a different amount from the expected range.

7. The data it would generate

Bench	Under trees	Hours too hot (July)	Hours wet (October)	Hours to dry after rain	% sittable, 2025	% sittable, 2050s
Bench 1	No	142	61	3	81%	74%
Bench 2	Yes	9	188	11	76%	75%
Bench 3	No	120	70	4	82%	76%
Bench 4	Yes	4	230	14	72%	72%

Values show the format, not results.

- **Chart:** the share of benches sittable each week from June to November. One line for 2025, one for the 2050s with a band showing the range across the 100 runs.
- **Map:** my Assignment 2 park map with each bench as a dot split in two: one half colored by hours too hot in July, the other by hours wet in October.

8. What it can't see

- Real bench materials and paint colors, which change how hot they get. Most OSM benches have no material listed.
- Shade from buildings, and leaves falling in autumn. The tree cover map is from 2017 and doesn't change.
- Puddles and drainage on individual benches. Each bench is treated as one surface.
- Weather differences within the park. One station stands in for the whole park.
- People sitting on benches, which can warm or dry them.
- How comfortable the sitter feels, which also depends on air temperature and humidity, not just the bench.